

VIREO AUDIO — SUPPORT SLA ANALYST

Project Understanding & Execution Guide — Revised / Verified

Version 2 — corrected after re-checking the supplied policy, email thread, data dictionary and CSV data

1. Executive Summary

Vireo Audio is evaluating us as a vendor for a support-operations analytics task. The client wants a weekly first-response SLA breach report by agent and shift so the Support Operations Manager has reliable numbers for follow-up conversations. The assignment deliberately gives more material than can comfortably be implemented in five hours; scope selection and the ability to explain decisions are part of the evaluation.

The core product is therefore a small, reproducible analytical tool—not an elaborate AI system. Its foundation must be deterministic: clean the ticket data, handle migration duplicates, interpret timestamps correctly, map agents to historical roster assignments, apply the channel-specific SLA rules, flag breaches, and quantify the ₹350 SLA-credit impact. AI should be used only where it adds value to free-text interpretation or manager-facing explanation.

2. The Client's Actual Ask

Neha Kulkarni's explicit request is: a breach report by agent and shift, weekly. She wants the numbers available for conversations with the right people. The final email repeats the requirement and says 'Weekly. Nothing fancy.'

The assignment does not provide a complete specification. If something remains ambiguous, we are expected to decide, document the decision, and explain why.

3. What SLA Means Here

SLA = Service Level Agreement. In this project, the relevant SLA is the first-response service level: the first reply by a human agent measured from ticket creation.

Channel	Target	Breach condition
Chat	15 minutes	First human response is later than 15 minutes
Voice callback	2 hours	First human response/callback is later than 2 hours
Social	4 hours	First human response is later than 4 hours
Email	8 hours	First human response is later than 8 hours

The policy says every ticket that misses its first-response target automatically issues a ₹350 store credit on resolution, charged to the SLA credit line in the support P&L. Breaches are reported against the resolving agent.

4. What Was Supplied

Resource	Verified purpose	How we use it
tickets.csv	18 months of support tickets; UTC timestamps; channel, agent, status, text and other fields	Primary event-level analytical data
agents.csv	One row per roster assignment with agent, site, team, shift, tier and from/to dates	Historical organizational context
support-policy.pdf	SLA targets, breach credit, shift definitions, teams/tiers, timestamp rules and reporting definitions	Business-rule source of truth
email-thread.txt	Conversation already held between Vireo and our side	Operational context and stakeholder constraints
orders.csv	Order references, values and manufacturing lot	Optional context
customers.csv	Customer profile and Care Plus information	Optional context
products.csv	Product family, cost, retail price and warranty period	Optional context / business-case support
README / data-pack document	Column definitions and join guidance	Data dictionary

The supplied data-pack document explicitly says the tickets export covers 1 Jan 2025–30 Jun 2026, timestamps are UTC, agent_id is the resolving agent, and agents.csv is one row per assignment. It also states that customer_id + product_sku is the fallback join for product context.

5. Resource Relationships

Core relationship:

tickets.csv → agent_id → agents.csv → historical shift / site / team / tier

tickets.csv → channel → support-policy.pdf → SLA target → breach flag

breach flag → support-policy.pdf → ₹350 SLA-credit impact

tickets.csv → customer_id / order_id / product_sku → reference CSVs → optional context

email thread → UTC/IST warning + June movement + duplicate warning + headcount constraint + presentation concern

6. Policy Rules That Directly Affect Our Tool

- Chat is available 24×7; overnight coverage is provided by the Indore Night shift.
- Email is accepted 24×7 and worked in queue order.
- Voice IVR takes callback requests 08:00–22:00 IST.
- Social is worked by the Chat Frontline team.

- First response means the first reply by a human agent.
- SLA targets are 15 min / 2 hr / 4 hr / 8 hr for chat / voice / social / email.
- Each missed first-response target generates a ₹350 store credit on resolution.
- Breaches are reported against the resolving agent.
- Shift definitions are IST: Morning 06:00–14:00, Day 14:00–22:00, Night 22:00–06:00.
- Roster rows represent assignment periods. Agents can move shift/site and retain the same agent_id.
- Queue coverage outside staffed hours is picked up by the next shift in order of creation.
- Tier 2 work is multi-touch and measured on resolution in days; Tier 2 agents must not be compared with Tier 1 on volume metrics.
- Blank CSAT means no response and must not be treated as zero. Legacy rows use 0 for no response.
- The current helpdesk went live on 14 Sep 2025. Earlier tickets were migrated from Freshdesk; some were re-imported under both systems.
- API timestamps are UTC; standard helpdesk reports use IST.

7. What the Email Thread Adds

- Neha wants a weekly breach report by agent and shift.
- Sameer explicitly warns that the API export is UTC while policy shifts are IST.
- Sameer explicitly warns that roster assignments change over time and some people moved in June.
- Sameer explicitly warns that some tickets appear twice because of migration re-import.
- Sameer notes that legacy CSAT uses 0 where the new system uses blank.
- About forty messages are failed IVR transcripts.
- Finance says headcount is frozen until Q4; 'hire more people' cannot be the proposed answer.
- Priya says the Morning chat team is demoralised and asks us not to build something that unfairly makes the situation worse.
- Neha nevertheless confirms the requirement: report by agent and shift, weekly, with nothing fancy.

8. Data Quality / Engineering Problems

Issue	Why it matters	Our treatment
Duplicate ticket IDs from migration re-import	Inflates volume and breaches if counted twice	Deduplicate to one canonical row per ticket_id using a documented rule
UTC timestamps vs IST shifts	Can place an event in the wrong shift/day	Parse as UTC and derive IST reporting fields
Historical roster assignments	Current roster is not valid for all historical tickets	Use time-valid roster mapping
No assigned_at field	We cannot observe the exact handoff/assignment timestamp	Document the roster-date assumption explicitly
Legacy CSAT 0	0 means no response in legacy data	Exclude legacy no-response values from CSAT averages
Tier 2 vs Tier 1	Different work model	Do not compare Tier 2 with Tier 1 on ticket-volume metrics
Failed IVR transcripts	Text can be low quality	Do not make deterministic SLA calculations depend on text quality

9. A Critical Implementation Assumption

The data pack gives us agent_id and roster from/to dates, but it does not provide an assigned_at/event timestamp. The policy says the report is against the resolving agent, while the ticket file defines agent_id as the agent who resolved the ticket. For a reproducible first version, we should time-match the agent's roster assignment using the ticket creation timestamp because it is available for every ticket and is the only common event timestamp that lets us assign a roster row consistently across open/pending records as well.

This is an assumption, not a fact supplied by Vireo. If the final implementation uses this rule, it must be stated in the README and validation section. Tickets that span a roster change are a known limitation because the dataset does not expose the actual assignment/handoff timestamp.

10. Deterministic SLA Logic

For each unique ticket:

response_minutes = first_response_at – created_at
sla_minutes = channel-specific target
breach = response_minutes > sla_minutes

The policy defines a breach as first response later than the target, so a response exactly at the target is not a breach.

The core engine does not need an LLM. Python/Pandas is more appropriate for timestamp arithmetic, joins, deduplication and aggregation.

11. Verified Data Counts

A fresh check of the supplied CSVs gives the following base numbers before any optional AI text analysis:

Metric	Verified value
Raw ticket rows	11,816
Unique ticket IDs	11,200
Duplicate rows removed by ticket_id	616
Unique-ticket SLA breaches	2,440
Overall breach rate	21.7857% (~21.8%)
SLA-credit exposure at ₹350/breach	₹854,000
Tier 1 tickets after roster mapping	10,448
Tier 1 breaches	2,265
Tier 1 breach rate	21.6788%
Tier 1 Morning tickets	5,803
Tier 1 Morning breaches	1,866
Tier 1 Morning breach rate	32.1558%
Morning breaches created overnight	1,452
Share of Morning breaches created overnight	77.8135%
Morning tickets created in normal hours: breach rate	11.0224%
Morning tickets created overnight: breach rate	70.9331%

Important: these figures use ticket_id deduplication, the channel SLA rules, UTC timestamps converted to IST for time-of-day analysis, and the documented creation-time roster assumption described above. They are our computed analysis, not numbers explicitly stated by the client.

12. The Main Operational Signal

The strongest current signal is not simply 'Morning agents are bad.' Tier 1 Morning has a 32.16% breach rate, but 77.81% of its breach tickets were created during the 22:00–06:00 IST overnight period. Morning tickets created during normal hours have a much lower observed breach rate (11.02%), while overnight-created Morning tickets have a 70.93% breach rate.

This supports an operational question around overnight queue accumulation and handoff/coverage. It does not prove that Morning staffing or the June reshuffle caused the breaches. The final report must distinguish observed correlation from causation.

13. June Reshuffle — What We Can Safely Say

The roster shows three Indore Chat Frontline agents moving from Night to Day effective 30 June 2025. The email thread also explicitly says some people moved in June and that the Morning team is under pressure.

A before/after comparison can be useful, but it must be presented as a diagnostic signal rather than a causal claim. Other factors can change at the same time, and the supplied data does not provide a controlled experiment.

Therefore the correct wording is: 'The breach pattern changes around the June roster boundary and is concentrated in overnight-created tickets.' Avoid: 'The June reshuffle caused the breach increase.'

14. Business Objective

PRIMARY OBJECTIVE: Build a small AI-assisted weekly SLA analysis tool that reliably identifies first-response breaches by agent and shift, quantifies their financial impact, and provides operational context so Vireo Support Operations can investigate and act on the main sources of SLA loss.

The assignment gives 'cut X from 22% to 15%, worth about ₹Y a quarter' as the shape of the requested business goal. It does not explicitly mandate 15% as Vireo's official target. Therefore, we should treat 15% as a proposed project target for the business case, not as an established client commitment.

At roughly 650 tickets/week, the observed 21.7857% rate corresponds to about 141.6 breaches/week. At ₹350 each, that is about ₹49.6k/week or ₹645k over a 13-week quarter. At 15%, it would be about 97.5 breaches/week and ₹34.1k/week. The difference is approximately ₹15.4k/week or ₹201k per quarter in potential avoided SLA credits, assuming the same ticket volume and mix.

15. What the Tool Must Answer

1. What is the overall first-response breach rate?
2. How does the breach rate change week by week?
3. Which shifts have the highest breach volume and breach rate?
4. Which agents are associated with the most breaches?
5. Which channels contribute most to the problem?
6. What is the associated SLA-credit exposure?
7. How much of the Morning problem is linked to overnight-created tickets?
8. What changes around the June roster boundary?

- 9. What recurring text themes appear in a checked sample of breach tickets?
- 10. What should the Support Operations Manager investigate next?

16. Where AI Should Be Used

AI is useful for the unstructured parts of the project, not the accounting-like parts.

Use AI for	Do not use AI for
Classifying sampled breach messages into operational themes	Calculating response duration
Summarising recurring customer/agent-note themes	Determining whether a ticket breached
Producing an evidence-grounded manager summary from validated metrics	Deduplicating ticket IDs
Helping explain text-heavy failure cases	Historical roster joins

The AI output must be evaluated separately. It should never be allowed to invent breach counts, costs, shifts or other deterministic facts.

17. Recommended Technical Architecture

Recommended stack: Python + Pandas for the analytical engine, Streamlit for the lightweight interface, and one controlled LLM/API path for text analysis or manager-facing explanation.

Pipeline:

CSV inputs → validation → ticket deduplication → timestamp parsing → IST reporting fields → historical roster mapping → SLA calculation → weekly aggregation → operational analysis → optional AI text analysis → Streamlit report/export.

Do not add LangChain, LangGraph, RAG, a vector database, multi-agent orchestration or predictive ML unless a concrete requirement emerges. They are not needed to answer the client's core question within the five-hour cap.

18. Suggested Tool Screens

- Top KPIs: breach rate, breach count, SLA-credit exposure.
- Weekly breach trend.
- Shift breakdown: Morning / Day / Night.
- Agent table: tickets, breaches, breach rate, tier and relevant context.
- Channel breakdown.
- Overnight-created ticket analysis.
- June roster-boundary comparison.
- Optional AI explanation of selected patterns.
- Downloadable CSV/report.

19. Validation — How We Prove It Works

- 11. Confirm raw row count and unique ticket_id count.
- 12. Verify duplicate handling on known duplicate ticket IDs.

13. Manually calculate a sample of response times and compare with code.
14. Test exact SLA boundary cases: exactly at target and just after target.
15. Test all four channels.
16. Test UTC-to-IST conversion around 06:00 and 22:00 shift boundaries.
17. Verify roster joins and explicitly report unmatched/ambiguous cases.
18. Check that Tier 2 is separated from Tier 1 volume comparisons.
19. For AI text classification, manually label a sample and report sample size and error rate.
20. Document the cases the tool is known to get wrong.

20. Scope We Should Build

- Reliable data/SLA engine.
- Duplicate-safe ticket population.
- Historical roster mapping with a documented assumption.
- Weekly agent/shift/channel metrics.
- Financial impact calculation.
- Overnight operational-context analysis.
- One small Streamlit tool.
- One focused AI capability.
- Validation tests and a known-limitations section.
- README, memo, screen recording and submission form.

21. Scope We Should Leave Out

- Predictive model for future SLA breaches.
- Customer churn or CSAT prediction.
- Large RAG/vector database architecture.
- Multi-agent system.
- Automated staffing optimiser.
- Product recommendation.
- Deep order/customer analytics that do not affect the SLA question.
- Polished cloud infrastructure if a clean local run is sufficient.

The assignment explicitly rewards good scope decisions. Leaving something out is acceptable when we can explain why it does not materially move the client's stated objective within the five-hour cap.

22. Final Deliverables

Deliverable	Acceptance requirement
Working AI-assisted tool	Starts from README on a clean machine
Business goal	Specific number and money impact
Evidence it works	Sample size, checking method, error rate and failure type
One-page memo	Non-technical; maximum 11 minutes reading
Screen recording	Maximum 3 minutes; prompts, changes and discarded approaches; no slides
submission-form.md	Every question completed

Public GitHub repo	Runnable project and documentation
Public Google Drive recording link	Included in submission form

23. Submission Form — What We Need to Answer

- What did we build, and what business outcome does it move?
- What does one run cost, and what would a month cost at roughly 650 tickets/week? Show arithmetic.
- How do we know it works? Sample size, checking method, error rate and wrong-case type.
- Did we change, narrow or push back on the client's ask? What, when and why?
- What is wrong with what we are handing over? Specific bugs, shortcuts and known gaps.
- What did we deliberately leave out, and why?
- Anything useful we built/found beyond the ask?
- What AI tools/models were used, where they helped, where they wasted time and what was discarded?
- Three-minute recording link.
- Three things a Monday handoff owner needs to know.
- Honest hours spent.
- Public GitHub repository link.

24. Important Cost Distinction

The ₹350 SLA credit is the business cost of a breach. It is not automatically the cost of running our tool. Tool-run cost depends on whether we use a paid LLM/API and how many calls/tokens are used. If we implement the deterministic analytics locally and use no paid model calls, the run cost can be stated as ₹0 for model/API usage, with local compute treated separately. If we use a paid model, the submission must show the actual call/token arithmetic.

25. Known Limitations We Should Be Honest About

- The 15% target is a proposed business-case target, not an explicit Vireo policy target.
- The ₹201k/quarter estimate is a scenario based on 650 tickets/week, the observed breach rate, and ₹350 per breach; it is not a guaranteed saving.
- Roster matching uses ticket creation time because the supplied data does not expose assigned_at. This is an explicit analytical assumption.
- A ticket can span a roster change; without assignment events, exact historical shift attribution for such tickets cannot be proven.
- The June pattern is correlation/diagnostic evidence, not proof of causation.
- AI text classification can be wrong, particularly on failed IVR transcripts.
- A breach attributed to a resolving agent can reflect upstream queue/coverage conditions, not necessarily individual performance.
- The tool is an analytical aid; it does not automatically change staffing, queues or support policy.

26. End-to-End Execution Plan

21. Formalise the business and policy rules.
22. Inspect all columns and data types.

23. Profile missing values, duplicate IDs and date ranges.
24. Define and test the canonical ticket deduplication rule.
25. Parse timestamps as UTC.
26. Create IST date/hour/week fields.
27. Map agents to roster assignments using the documented timestamp assumption.
28. Calculate first-response duration and channel SLA.
29. Create breach flag and ₹350 impact.
30. Aggregate weekly by shift, agent, tier and channel.
31. Investigate overnight-created tickets and the June roster boundary.
32. Select a small manually checked sample for text analysis.
33. Add the focused AI layer.
34. Build the Streamlit interface.
35. Run validation tests and document error cases.
36. Write README and one-page memo.
37. Complete submission form and screen recording.
38. Run the project from a clean environment before submission.

27. The Project in One Sentence

Build a reliable weekly first-response SLA analyst for Vireo Audio that combines ticket data, historical roster context and policy rules to identify breaches, quantify their ₹350-per-breach impact, and use AI selectively to explain operational patterns—especially where overnight-created work is associated with Morning-shift breaches.

28. Final Mental Model

Think of the project as five layers:

Layer	Question
1. Data	What actually happened in each ticket?
2. Rules	Was the response inside the correct SLA?
3. Organisation	Which historical agent/shift context applies?
4. Business	How large is the problem and what does it cost?
5. AI / insight	What recurring text or operational pattern helps explain it?

If Layers 1–4 are wrong, AI cannot rescue the project. Therefore our implementation order should always be data correctness → SLA correctness → roster correctness → business analysis → AI interpretation → presentation.

29. Verification Note

This revised version was re-checked against the supplied email thread, support-policy PDF, data-pack/README document, tickets.csv and agents.csv. The numerical ticket/breach figures were recalculated from the supplied CSVs. Where the source does not establish a fact—for example an official 15% target or an assigned_at timestamp—the document labels the item as an assumption, proposed target or limitation instead of presenting it as client fact.